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Use of a basic oxygen furnace to produce granulated metallic units, and associated systems, devices, and methods

Abstract

Systems and methods for using a liquid hot metal processing unit to produce granulated metallic units (GMUs) are disclosed herein. In some embodiments of the present technology, a liquid hot metal processing system for producing GMUs comprises a liquid hot metal processing unit including a granulator unit. The granulator unit can include a tilter positioned to receive and tilt a ladle, a controller operably coupled to the tilter to control tilting of the ladle, a tundish positioned to receive the molten metallics from the ladle, and a reactor positioned to receive the molten metallics from the tundish. The reactor can be configured to cool the molten metallics to form granulated metallic units (GMUs).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2052783	12/1935	Mart	N/A	N/A
3316075	12/1966	Grady	65/159	F27D 3/15
3548919	12/1969	Gyongyos	164/434	B22D 11/064
3720404	12/1972	Carlson	75/375	C21C 5/30
3810542	12/1973	Gloster et al.	N/A	N/A
3888956	12/1974	Klint	N/A	N/A
3898037	12/1974	Lange et al.	N/A	N/A
3923935	12/1974	Cates	N/A	N/A
4139369	12/1978	Kandler et al.	N/A	N/A

4153965	12/1978	Merly	15/104.096	B22D 41/001
4294784	12/1980	Mailund	N/A	N/A
4330511	12/1981	Nelson et al.	N/A	N/A
4402884	12/1982	Koike et al.	N/A	N/A
4416707	12/1982	Foster et al.	N/A	N/A
4546907	12/1984	Kemble	N/A	N/A
4694886	12/1986	Sakaguchi	N/A	N/A
4786322	12/1987	Green	75/312	C21C 1/10
4893568	12/1989	Adams	N/A	N/A
5017218	12/1990	Lundström et al.	N/A	N/A
5084093	12/1991	Yamaoka et al.	N/A	N/A
5458671	12/1994	Butler et al.	N/A	N/A
5552058	12/1995	Fanning	N/A	N/A
5673779	12/1996	Spickelmire	N/A	N/A
6041906	12/1999	Howard	N/A	N/A
6287362	12/2000	Levey et al.	N/A	N/A
8646700	12/2013	Lundström et al.	N/A	N/A
9840746	12/2016	Lundström	N/A	N/A
10486234	12/2018	Lundström	N/A	N/A
10618112	12/2019	Lundström	N/A	N/A
2001/0002535	12/2000	Liebig et al.	N/A	N/A
2002/0026967	12/2001	Buenemann et al.	N/A	N/A
2003/0015315	12/2002	Atsushi et al.	N/A	N/A
2003/0164062	12/2002	Lundstrom et al.	N/A	N/A
2005/0133192	12/2004	Meszaros et al.	N/A	N/A
2009/0015028	12/2008	Zamorano	N/A	N/A
2009/0171512	12/2008	Duncan	N/A	N/A
2013/0180360	12/2012	Kim et al.	N/A	N/A
2015/0330725	12/2014	Gurney	N/A	N/A
2020/0122237	12/2019	Kemper	N/A	C21C 7/0087

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1847411	12/2005	CN	N/A
101301683	12/2007	CN	N/A
102700957	12/2011	CN	N/A
103661440	12/2015	CN	N/A
205496553	12/2015	CN	N/A
107385135	12/2016	CN	N/A
207205279	12/2017	CN	N/A
212049627	12/2019	CN	N/A
112305944	12/2020	CN	N/A
113549715	12/2020	CN	N/A
214470214	12/2020	CN	N/A
113828788	12/2020	CN	N/A
114433854	12/2021	CN	N/A
216709283	12/2021	CN	N/A
115417184	12/2021	CN	N/A

115889716	12/2022	CN	N/A
116550982	12/2022	CN	N/A
1287510	12/1971	GB	N/A
H02236212	12/1989	JP	N/A
H06212212	12/1993	JP	N/A
2001107118	12/2000	JP	N/A
2002282866	12/2001	JP	N/A
2005154847	12/2004	JP	N/A
2020164992	12/2019	JP	N/A
7033949	12/2021	JP	N/A
2022149432	12/2021	JP	N/A
20010111750	12/2000	KR	N/A
100370611	12/2002	KR	N/A
100847667	12/2007	KR	N/A
101086315	12/2010	KR	N/A
101091977	12/2010	KR	N/A
20130076110	12/2012	KR	N/A
20140102742	12/2013	KR	N/A
20140120631	12/2013	KR	N/A
20160034000	12/2015	KR	N/A
101649584	12/2015	KR	N/A
20230028951	12/2022	KR	N/A
2007105039	12/2006	WO	N/A

OTHER PUBLICATIONS

English translation of CN 112305944 (originally published Feb. 2, 2021), obtained from PE2E search. cited by examiner
English translation of JP 2022149432 (originally published Oct. 6, 2022), obtained from PE2E search. cited by examiner
“Granulation of Hot Metal,” by 360 Editor, Apr. 23, 2018, 4 pages. cited by applicant
Beskow et al., “Industrial and High-Capacity Production of Granulated Pig Iron,” Association for Iron & Steel Technology, AISTech, May 6-9, 2019, pp. 485-493. cited by applicant
Beskow et al., “World's First High-capacity Granshot © Iron Granulation in Operation at SSAB Oxelosund,” AISTech 2009 Proceedings—vol. I, pp. 211-218. cited by applicant
Dupon et al., “Granshot Iron Granulation for Optimized Plant Logistics,” METEC InSteelCon Jun. 27-Jul. 1, 2011, Düsseldorf, Germany; 7 pages. cited by applicant
Granulated Pig Iron (GPI), International Iron Metallics Association, accessed Sep. 17, 2024 from https://www.metallics.org/gpi.html; 4 pages. cited by applicant
Industrial and High Capacity Granulation of Pig Iron, Data Sheet, Uvån Hagfors Teknologi AB, Jun. 2019, 4 pages. cited by applicant
Love et al., “Energy Recovery in Granshot ©—Process,” KTH Industrial Engineering and Management, Stockholm, Sweden, Nov. 25, 2008, 70 pages. cited by applicant
Lundstrom et al., “Pig Iron Granulation at Iscor Saldanha Steel,” AISTech 2004 Proceedings—vol. I, pp. 517-524. cited by applicant
Lundstrom, P. “Iron Granulation in Integrated Steel Plants,” Nordic Steel & Mining Review 2006, pp. 16-17. cited by applicant
Polanco et al., “Granulation Methods for Metals and Ferroalloys,” pp. 401-410. In: 72nd ABM Annual Congress, São Paulo, 2017. cited by applicant
Vesterberg et al., “Granulated metal product from direct tapped furnace—experience from operation at BEFESA Sweden,” ATS-JSI Conference, Dec. 17-18, 2012, Paris, France, 2 pages.

cited by applicant
International Iron Metallics Association (IIMA), "Use of Granulated Pig Iron (GPI) in the Electric Arc Furnace (EAF)," downloaded Nov. 30, 2024, dated 2018 (no month) (two pages) [wayback machine date of Feb. 9, 2022]. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2024/046275; Date of Mailing: Dec. 27, 2024; 11 pages. cited by applicant
Italimpianti Orafi, "Atomized metal powder: the revolution of the precious metals market," Online article, May 20, 2021; 2 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) The present application claims the benefit of U.S. Provisional Patent Application No. 63/581,946, filed Sep. 11, 2023, and titled "SYSTEM AND METHOD FOR CONTINUOUS GRANULATED PIG IRON (GPI) PRODUCTION," the disclosure of which is incorporated herein by reference in its entirety. The present application is related to the following applications, the disclosures of which are incorporated herein by reference in their entireties: U.S. patent application Ser. No. 18/882,116, filed Sep. 11, 2024, and titled "RAILCARS FOR TRANSPORTING GRANULATED METALLIC UNITS, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,045, filed Sep. 11, 2024, and titled "LOADING GRANULATED METALLIC UNITS INTO RAILCARS, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,191, filed Sep. 11, 2024, and titled "LOW-SULFUR GRANULATED METALLIC UNITS, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,638, filed Sep. 11, 2024, and titled "CONTINUOUS GRANULATED METALLIC UNITS PRODUCTION, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,256, filed Sep. 11, 2024, and titled "LOW-CARBON GRANULATED METALLIC UNITS, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,531, filed Sep. 11, 2024, and titled "TORPEDO CARS FOR USE WITH GRANULATED METALLIC UNIT PRODUCTION, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,384, filed Sep. 11, 2024, and titled "TREATING COOLING WATER IN IRON PRODUCTION FACILITIES, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,465, filed Sep. 11, 2024, and titled "USE OF RESIDUAL IRON WITHIN GRANULATED METALLIC UNIT PRODUCTION FACILITIES, AND ASSOCIATED SYSTEMS, DEVICES, AND METHODS"; U.S. patent application Ser. No. 18/882,501, filed Sep. 11, 2024, and titled "PROCESSING GRANULATED METALLIC UNITS WITHIN ELECTRIC ARC FURNACES, AND ASSOCIATED SYSTEMS AND METHODS".

TECHNICAL FIELD
(1) The present technology generally relates to converting a basic oxygen furnace facility to produce granulated metallic units, and associated systems, devices, and methods.
BACKGROUND
(2) Granulated pig iron (GPI) is a form of iron that is granulated into small, uniform particles, making it easier to handle, transport, and use in different metallurgical processes compared to conventional iron. The demand for GPI has been steadily increasing due to its versatile applications

in various industries, including automotive, construction, and manufacturing. The growing popularity of GPI can be attributed to its high purity, consistent quality, and the efficiency it brings to the production of steel and other iron-based products.
(3) Granulated pig iron is produced by rapidly cooling molten iron with water, resulting in the formation of granules. This process, known as granulation, is typically carried out after blast furnaces. However, current production methods are often characterized by intermittent production cycles due to various operational constraints, such as the need for periodic maintenance, fluctuations in raw material supply, and energy consumption issues. These interruptions not only affect the overall efficiency but also lead to increased production costs and variability in product quality. Therefore, there is a need for an improved production process that can ensure continuous and stable granulation of iron, thereby enhancing productivity and reducing operational costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following drawings.
(2) FIG. **1** is a schematic block diagram of a liquid hot metal processing system configured in accordance with embodiments of the present technology.
(3) FIG. **2** is a plan view of the liquid hot metal processing system of FIG. **1**, configured in accordance with embodiments of the present technology.
(4) FIG. **3** is a partially schematic, side cross-sectional view of a basic oxygen furnace vessel configured in accordance with embodiments of the present technology.
(5) FIG. **4** is a schematic plan view of two granulator units in a first arrangement and configured in accordance with embodiments of the present technology.
(6) FIG. **5** is a schematic plan view of two granulator units in a second arrangement and configured in accordance with embodiments of the present technology.
(7) FIGS. **6**A-**6**C are partially schematic side views illustrating tilting of a transfer vessel in accordance with embodiments of the present technology.
(8) FIGS. **7**, **8**A, and **8**B are front perspective, rear perspective, and side cross-sectional views, respectively, of a tundish configured in accordance with embodiments of the present technology.
(9) FIG. **9** is a perspective view of a stopper rod assembly configured in accordance with embodiments of the present technology.
(10) FIG. **10** is a perspective view of a tundish and a stopper rod assembly assembled together and configured in accordance with embodiments of the present technology.
(11) FIG. **11** is a schematic cross-sectional view of a granulation reactor configured in accordance with embodiments of the present technology.
(12) FIGS. **12**A and **12**B are schematic side and enlarged, side cross-sectional views, respectively, of an ejector configured in accordance with embodiments of the present technology.
(13) FIG. **13** is a schematic side view of a dewatering assembly configured in accordance with embodiments of the present technology.
(14) FIG. **14** is a schematic side view of a classifier assembly configured in accordance with embodiments of the present technology.
(15) FIG. **15** is a schematic process flow diagram illustrating granulation of iron in accordance with embodiments of the present technology.
(16) FIG. **16** is a flowchart illustrating a method for producing GMUs in accordance with embodiments of the present technology.
(17) A person skilled in the relevant art will understand that the features shown in the drawings are for purposes of illustrations, and variations, including different and/or additional features and

arrangements thereof, are possible.

DETAILED DESCRIPTION

I. Overview

(18) The present technology is generally directed to systems, devices, and methods for converting or retrofitting a basic oxygen furnace (BOF) facility or a liquid hot metal processing unit to produce granulated metallic units (GMUs). GMUs can be produced by forming molten iron in a blast furnace and rapidly cooling the molten iron with water to form granules. However, producing GMUs can require numerous large and complex equipment such as flow control devices, overhead cranes, ladles, lances, etc. To build a new GMU production facility from scratch can be costly.

(19) Embodiments of the present technology address at least some of the above described issues by converting or retrofitting a BOF facility to produce GMUs so that many of the existing equipment in the BOF facility can be repurposed. As described herein, some embodiments of the present technology can include a liquid hot metal processing system for producing granulated metallic units comprising a liquid hot metal processing unit including a ladle, a granulator unit, and an overhead crane. The ladle can be shaped to receive and store molten iron therein. The granulator unit can include a tilter positioned to receive and tilt the ladle, a controller operably coupled to the tilter to control tilting of the ladle, a tundish positioned to receive the molten iron from the ladle, and a reactor positioned to receive the molten iron from the tundish. The reactor can be configured to cool the molten iron to form GMUs. The overhead crane can be configured to transfer the ladle to and from the tilter.

(20) Specific details of several embodiments of the technology are described below with reference to FIGS. **1-16**. Other details describing well-known structures and systems often associated with furnaces, rails, conveyor belts, emission hoods, automated control systems, etc. have not been set forth in the following disclosure to avoid unnecessarily obscuring the description of the various embodiments of the technology. Many of the details, dimensions, angles, and other features shown in the Figures are merely illustrative of particular embodiments of the technology. Accordingly, other embodiments can have other details, dimensions, angles, and features without departing from the spirit or scope of the present technology. A person of ordinary skill in the art, therefore, will accordingly understand that the technology may have other embodiments with additional elements, or the technology may have other embodiments without several of the features shown and described below with reference to FIGS. **1-16**.

II. Embodiments of a Liquid Hot Metal Processing System

(21) FIG. **1** is a schematic block diagram of a liquid hot metal processing production system **100** (“the system **100**”) configured in accordance with embodiments of the present technology. As explained elsewhere herein, granulated metallic units (GMUs) can include granulated pig iron (GPI) or granulated steel (GS). Relatedly, molten metallics can include molten pig iron or molten steel. In some embodiments, the system **100** comprises a continuous system that can operate under continuous operations cycles, including in batch or semi-batch operations, for at least 2 hours, 4 hours, 6 hours, 8 hours, 10 hours, 12 hours, 16 hours, 20 hours, or 24 hours. The duration of the continuous operations cycles can depend at least in part on the size of the GMU to be produced by the system **100**. In some embodiments, the system **100** can produce at least 2,000 tons, 4,000 tons, 6,000 tons, 8,000 tons, 10,000 tons, or more of GMUs per hour. In some embodiments, the system **100** is configured to primarily produce GMUs.

(22) The system **100** can include a liquid hot metal processing unit **104** and a blast furnace **102** located outside of the liquid hot metal processing unit **104**. The blast furnace **102** can receive input materials (e.g., iron ore, coke, limestone, and/or preheated air) and/or recycled material, which can be sourced from downstream components of the system **100** as described in further detail herein. Equations (1)-(6) below detail some of the chemical processes controlled at the blast furnace **102**.

(23) $C + O_2$ .fwdarw. CO_2 (1) $CO_2 + C$ .fwdarw. 2CO (2)

Fe_2O_3 .fwdarw. 3CO .fwdarw. 2Fe + $3CO_2$ (3) Fe_2O_3 .fwdarw. 3C .fwdarw. 2Fe + 3CO (4)

$CaCO_3$ .fwdarw. CaO + CO_2 (5) CaO + SiO_2 .fwdarw. $CaSiO_3$ (6)

(24) Equation (1) represents the combustion of coke, which is a form of carbon. When coke reacts with oxygen gas introduced into the furnace (e.g., via an oxygen lance), it forms carbon dioxide. This exothermic reaction releases a significant amount of heat, which is essential for maintaining the high temperatures required for subsequent reactions. The carbon dioxide produced via Equation (1) further reacts with additional coke to form carbon monoxide, as illustrated by Equation (2). This endothermic reaction helps to moderate the temperature within the blast furnace **102**. Equations (3) and (4) represent the reduction of iron ore (Fe.sub.2O.sub.3). As illustrated by Equation (3), the iron oxide reacts with the carbon monoxide produced via Equation (2), which acts as a reducing agent to convert iron ore into iron and produces carbon dioxide as a byproduct. Alternatively, as illustrated by Equation (4), the iron ore may be reduced directly by the coke, albeit less commonly. Equations (5) and (6) represent the formation of slag. As illustrated by Equation (5), the calcium carbonate/limestone (CaCO.sub.3) can decompose into calcium oxide and carbon dioxide at the high temperatures of the blast furnace **102**. As illustrated by Equation (6), the calcium oxide can then react with silica (SiO.sub.2), an impurity in the iron ore, to form calcium silicate (CaSiO.sub.3), also known as slag. The blast furnace **102** can output molten metallics (from Equations (3) and (4)) and slag (from Equations (5) and (6)).
(25) The liquid hot metal processing unit **104** can include a scrap yard **106**, a charging aisle **108**, a furnace aisle **110**, and a teeming aisle **120**. The charging aisle **108** can provide space for a transfer vessel **122** (e.g., a torpedo car, a ladle, etc.) holding molten metallics from the blast furnace **102**. The furnace aisle **110** can include one or more BOF vessels **112**. The scrap yard **106** can serve as a stockpile area for scraps that may be fed into the BOF vessels **112**. The teeming aisle **120** can include additional transfer vessels **122** that can receive the output of the BOF vessels **112**. Returning to the blast furnace **102**, the transfer vessel **122** can transfer the molten metallics from the blast furnace **102** to the liquid hot metal processing unit **104** via either Path A or Path B.
(26) Path A involves transporting the molten metallics to a desulfurization unit **140** in the liquid hot metal processing unit **104** directly or via the charging aisle **108** via the transfer vessel **122**. The desulfurization unit **140** can include equipment to reduce a sulfur content of the molten metal. For example, one or more lances can be used to deliver magnesium (Mg), calcium carbide (CaC.sub.2), or other sulfur-reducing agent to the molten metallics. In some embodiments, the molten metallics is desulfurized while remaining inside the transfer vessels **122**. In some embodiments, the molten metallics is desulfurized in a torpedo car, and the desulfurized molten metallics is subsequently transferred to a ladle. Equations (7) and (8) below detail the reactions between the sulfur and the sulfur-reducing agents.
(27) Mg + S .fwdarw. MgS (7) CaC_2 + S .fwdarw. CaS + 2C (8)
(28) The resulting substances, including magnesium sulfide (MgS) and calcium sulfide (CaS), are not soluble in molten metallics and will therefore be in solid form (e.g., as solid particles) that can be more readily removed at the desulfurization unit **140** and/or further downstream. As discussed further herein, reducing the sulfur content can increase the quality of the GMUs product and/or allow the production process to be continuous. Notably, conventional facilities may not include a desulfurization unit or may otherwise lack the ability to desulfurize molten metallics. One reason for this is that conventional steelmaking facilities directly feed molten metallics from blast furnaces to basic oxygen furnaces, and opt to granulate the molten metallics only when the basic oxygen furnaces are down. Because producing GMUs is a backup operation for such facilities, the added complexity and costs associated with establishing desulfurization equipment may not be economical.
(29) After the desulfurization process, the desulfurized molten metallics can be fed to basic oxygen furnace (BOF) vessels **112** in the furnace aisle **110**. The BOF vessels **112** can reheat and/or reduce a

carbon content of the molten metal. For example, one or more oxygen lances can be used to deliver oxygen gas to the molten metallics in the BOF vessels **112**. The oxygen can react with carbon present in the molten metallics, causing combustion that can reheat the molten metallics, which may have cooled down to below a desired temperature range, such as between, 2000-3000° F., between 2300-2500° F., between 2300-2400° F., or between 2340-2350° F., or the iron liquidus/carbon equilibrium/eutectic point. Additionally or alternatively, the desired temperature can be at least 100° F., 200° F., 300° F., or 400° F. above a solidification temperature, depending on a chemical makeup of the composition (e.g., a silicon content of the composition). The reaction between the oxygen and the carbon can also reduce the carbon content of the molten metal, which can be desirable for tuning material properties of the end product. For example, the resulting GMUs may have a sufficiently low carbon content to be more steel-like. The molten metallics can subsequently be transferred from the BOF vessels **112** to the transfer vessels **122** in the teeming aisle **120**. On the other hand, Path B involves skipping the desulfurization unit **140** and the BOF vessels **112** and instead transferring the molten metallics from the blast furnace **102** directly to the transfer vessels **122** in the teeming aisle **120**.

(30) As shown in FIG. **1**, the liquid hot metal processing unit **104** can further include an overhead crane **130**, one or more granulator units **150**, and a cooling system **160**. The overhead crane **130** can be controlled (e.g., using a controller **105**) to carry the transfer vessels **122** (e.g., ladles) with the molten metallics to downstream units in the liquid hot metal processing unit **104**. In some embodiments, the overhead crane **130** carries the transfer vessels **122** to the one or more granulator units **150**. The system **100** can include one, two, three, four, five, six, or more granulator units **150**. Each granulator unit **150** can include a granulation reactor that receives and granulates molten metallics to form granulated products. For example, the granulation reactor can include a cavity that holds water and/or additives that help achieve a desired heat capacity and/or thermal conductivity, and the molten metallics can be transferred (e.g., poured, sprayed) onto a target of the reactor holding the water. The water can be maintained at a sufficiently low temperature by the cooling system **160** (e.g., cooled directly by pumping the water between the granulator units **150** and the cooling system **160**, cooled indirectly by pumping a coolant separate from the water that receives the molten metallics). In some embodiments, each granulator unit **150** includes one or more components for controlling the flow of molten metallics from the transfer vessels **122** to the granulation reactor. As one of ordinary skill in the art will appreciate, flow control can affect the shape, size, and quality of the granulated products. The granulator units **150** can also include a dewatering assembly for drying the granulated products from the granulation reactor to output GMUs. The granulator units **150** can further include a classifier assembly for filtering the filtrate from the dewatering assembly to output fines.

(31) The liquid hot metal processing unit **104** can further include a product handing unit **170** to receive the GMUs output by the granulator units **150** (e.g., by the dewatering assembly), and a fines handling unit **172** to receive the GMU fines output by the granulator units **150** (e.g., by the classifier assembly). In some embodiments, the product handling unit **170** and/or the fines handling unit **172** each includes one or more conveyor belts, diverters, stockpile locations, etc. The liquid hot metal processing unit **104** can further include a transfer vessel preparation unit **180** that can remove slag and/or kish from the transfer vessel **122**. For example, after delivering the molten metallics to the granulator units **150**, the overhead crane **130** can carry the transfer vessels **122** to the transfer vessel preparation unit **180** to be cleaned or otherwise prepared for the next cycle of transferring molten metallics.

(32) The system **100** can additionally include a product loadout **171**, a fines loadout **173**, slag processing **182**, and a scrap storage **184** outside the liquid hot metal processing unit **104**. The product loadout **171** can be downstream of the product handling unit **170** to receive GMUs products. The fines loadout **173** can be downstream of the fines handling unit **172** to receive GMU fines. The slag processing **182** can be downstream of the transfer vessel preparation unit **180** to

receive slag removed from the transfer vessels. The scrap storage **184** can be downstream of the granulator units **150** to receive thin pig and/or iron skulls. As shown in FIG. **1**, the fines at the fines loadout **173**, slag and/or iron from the tundish at the granulator units **150**, and/or the thin pig and/or iron skulls at the scrap storage **184** can be fed back into the blast furnace **102** as recycled materials. In some embodiments, the recycled materials are processed (e.g., pelletized) prior to being fed into the blast furnace **102**.

(33) Furthermore, emissions from various components of the system **100** can be collected and directed towards one or more dust collection units **190** (e.g., one or more baghouses, one or more scrubbers, one or more precipitators, etc.) in the liquid hot metal processing unit **104**. For example, emissions from the desulfurization unit **140**, the granulator units **150**, and the transfer vessel preparation unit **180** can be collected via fume hoods and directed to the dust collection unit **190** via pipes. The dust collection unit **190** can filter the emissions to remove dust therefrom so that clean waste gas is sent to stacks (not shown) to be released into the atmosphere, and the removed dust can be directed to further processing **192**.

(34) FIG. **2** is a plan view of the BOF GMU system **100**, configured in accordance with embodiments of the present technology. As shown the blast furnace **102** can be removed from the liquid hot metal processing unit **104**. The liquid hot metal processing unit **104** can include the charging aisle **108**, the furnace aisle **110** adjacent to the charging aisle **108**, and the teeming aisle **120** adjacent to the furnace aisle **110**. The desulfurization unit **140** can be located away from the charging aisle **108**, and the granulator units **150** can be located adjacent to the charging aisle **108**. In the illustrated embodiment, the overhead crane **130** can travel along an L-shaped path extending next to the charging aisle **108**, the furnace aisle **110**, the teeming aisle **120**, and desulfurization unit **140**, and the granulator units **150**. Therefore, the overhead crane **130** can carry the transfer vessels **122** from the teeming aisle **120** to the desulfurization unit **140** and/or the granulator units **150**. Furthermore, the liquid hot metal processing unit **104** can include the scrap yard **106**, the one or more dust collection units **190**, a return water sump **202**, and an oxygen storage area **204**. The return water sump **202** can receive water collected from, e.g., the classifier included in the granulator units **150**. The oxygen storage area **204** can store oxygen for, e.g., the oxygen lances used at the BOF vessels **112**.

(35) Referring to FIGS. **1** and **2** together, converting a BOF facility or liquid hot metal processing unit to produce GMUs as opposed to, e.g., building a new GMU production facility can significant costs because many of the required equipment may be present at the BOF facility. Such existing equipment can include the overhead crane **130**, the desulfurization unit **140**, the transfer vessels **122**, and select components included in the granulator units **150**, as discussed in further detail herein. Accordingly, embodiments of the present technology are expected to enable the production of GMUs at a fraction of the cost compared to building a new production facility and initiate production much earlier due to time savings associated with constructing a new production facility.

(36) Also, the system **100** is expected to be able to continuously produce GMU, unlike conventional GMU production systems. First, the inclusion of the desulfurization unit **140** provides several advantages. For example, GMUs with lower sulfur content produces less slag when melted at an electric arc furnace downstream, saving associated time, costs, and energy consumption. For example, a relatively lower level of sulfur and/or carbon content can improve throughput and increase production of the downstream electric arc furnace (EAF) and/or ladle metallurgical furnace (LMF). The use of GMUs with lower sulfur content can also ease maintaining the desired chemical composition and temperature, reducing the frequency of adjustments and interruptions during the melting cycle. Lower sulfur levels can also result in less wear and tear on other components of the system, reducing maintenance needs and associated downtime.

(37) Second, the inclusion of a plurality of granulator units **150** allows molten metallics to be granulated at separate granulator units in parallel. The granulator units **150** can also serve as backups for one another in case one of the granulator units **150** is down (e.g., due to

malfunctioning components, maintenance, etc.) or in a turndown situation. Furthermore, in some embodiments, the various components of the granulator units **150** are modular. For example, each of the components can be easily and independently removed (e.g., for maintenance) and/or replaced (e.g., via an overhead crane) without impacting operation of the other components.
(38) As discussed above, the system **100** is designed for continuous operation. Relative to non-continuous GPI production systems, embodiments of the present technology enhance energy efficiency and reduces emissions by minimizing the need for frequent shutdowns and restarts, which are often associated with excessive venting and/or less efficient operations. As described herein, some embodiments include (i) a desulfurization unit that lowers the sulfur content in molten metal, thereby reducing sulfur dioxide (SO.sub.2) emissions, (ii) dust collection units that filter out particulate matter, thereby reducing air pollution, (iii) infrastructure to recycle fines, slag, iron skulls and other residual and/or previously-processed metallics, thereby reducing the environmental impact associated with raw material extraction and conserving natural resources, and/or (iv) water management and cooling systems that minimize heat losses, enhance thermal efficiency of production processes, and optimize water consumption. Overall, the continuous GMU production system **100** enhances productivity while minimizing greenhouse gas emissions and waste, contributing to more sustainable industrial practices and helping mitigate climate change.
(39) Relatedly, conventional iron production has a significant environmental impact due to its high energy consumption and emissions of pollutants. As such, embodiments of the present technology which relate to GMU production systems can reduce this impact. Sulfur, phosphorus, and silicon in GPI negatively affect the quality and properties of final metal products, leading to issues like reduced ductility, toughness, and weldability, as well as surface defects and brittleness. These impurities also contribute to the formation of non-metallic inclusions and excessive slag, complicating metal processing and compromising product quality. Sulfur, in particular, accelerates the wear and erosion of metal processing equipment, increasing maintenance costs and decreasing equipment lifespan. Embodiments of the present technology include methods for removing these impurities in part can improve the quality and durability of final metal products and enhances the efficiency and lifespan of processing equipment, leading to cost savings and more sustainable production practices.
(40) FIG. **3** is a partially schematic, side cross-sectional view of the BOF vessel **112** configured in accordance with embodiments of the present technology. As discussed above with reference to FIG. **1**, the BOF vessel **112** can be used if Path A is taken. The BOF vessel **112** can include a shell **302** (e.g., a steel shell) defining an interior chamber for receiving materials and a liner material **303** (e.g., a refractory liner) along the inner surface of the shell **302**. The shell **302** can include a tap hole **304** through which materials can be output by the BOF vessel **112**, and a fume collection hood or pipe **306** extending from the top portion of the shell **302**. As shown, one or more oxygen lances **308** can be inserted into the interior chamber of the BOF vessel **112**.
(41) In operation, the BOF vessel **112** can receive molten metallics **305** from, e.g., a transfer vessel such as a torpedo car coming from the blast furnace **102**. In some embodiments, molten slag **307** can be present on the surface of the molten metallics **305**. The oxygen lances **308** can be operated to deliver oxygen gas (e.g., from the oxygen storage area **204**) to the molten metallics **305** and the molten slag **307**. The oxygen gas can combust carbon remaining in the materials (e.g., leftover coke), thereby simultaneously heating the molten metallics **305** and reducing a carbon content thereof. The molten metallics **305** may have cooled while transported from the blast furnace **102** to the BOF vessel **112**, and the temperature of the molten metallics **305** can be raised to be within a desired temperature range that, e.g., avoids premature solidification of the molten metallics **305**. As one of ordinary skill in the art will appreciate, maintaining the molten metallics **305** in a fluid state can facilitate proper granulation downstream of the BOF vessel **112**. Emissions generated in the BOF vessel **112** can be directed to the dust collection unit **190** or other emissions unit via the fume collection hood or pipe **306**.

(42) FIG. **4** is a schematic plan view of two granulator units **450** in a first arrangement and configured in accordance with embodiments of the present technology. In the illustrated embodiment, a first granulator unit **450***a* and a second granulator unit **450***b* are positioned side-by-side. In the first arrangement, the first granulator unit **450***a* and the second granulator unit **450***b* can have their components arranged in an identical or generally similar layout. Each granulator unit **450** can include a tilter **410**, a tundish **420**, a granulation reactor **430**, an ejector **440**, a dewatering assembly **460**, and a classifier assembly **470**. The tilter **410** can be sized and positioned to receive one transfer vessel **122** (e.g., a ladle carried by the overhead crane **130**). The tundish **420** can be positioned to receive molten metallics from the transfer vessel **122** and supported directly above the granulation reactor **430**. The dewatering assembly **460** can be positioned next to the tilter **410** and the granulation reactor **430**. The ejector **440** can extend between the granulation reactor **430** and the dewatering assembly **460**. The classifier assembly **470** can be positioned next to the dewatering assembly **460**.
(43) Notably, many of the components of the granulator units **450**, such as the dewatering assembly **460** and the classifier assembly **470**, are positioned side-by-side as opposed to, e.g., on top of one another. This allows the overhead crane **130** to more easily access each of the components, allowing the components to be lifted and removed for maintenance and/or replacement. As discussed further herein, each of the components can include lift lugs that the overhead crane **130** can hook onto for lifting and moving. This layout can be in contrast to conventional facilities in which components are positioned over one another and thus difficult to quickly lift and remove using a crane.
(44) FIG. **5** is a schematic plan view of two granulator units **550** in a second arrangement and configured in accordance with embodiments of the present technology. In the illustrated embodiment, a first granulator unit **550***a* and a second granulator unit **550***b* are positioned side-by-side. Unlike in the first arrangement illustrated in FIG. **4**, in the second arrangement, the components of the first granulator unit **550***a* and the components of the second granulator unit **550***b* are not arranged in an identical or generally similar layout. Instead, in a generally mirrored arrangement, the tilters **410**, the tundishes **420**, and the granulation reactors **430** of the two granulator units **550** are positioned towards opposite ends and away from the same components of the other granulator unit **550**. The dewatering assemblies **460** and the classifier assemblies **470** are positioned between the tilters **410**, the tundishes **420**, and the granulation reactors **430** of the two granulator units **550**. Compared to the first arrangement of FIG. **4**, in the second arrangement of FIG. **5**, the two transfer vessels **122** can be positioned farther apart from one another, and the dewatering assemblies **460** can be positioned closer to one another and face each other so that, e.g., the GMUs products can be collected using a single container or conveyor.
(45) Referring to FIGS. **4** and **5** together, it will be appreciated that the illustrated embodiments merely show example arrangements or layouts of granulator units, and that the components thereof can be arranged differently. Moreover, in some embodiments, the system **100** may include only one granulator unit or three, four, five, six, or more granulator units.
(46) FIGS. **6**A-**6**C are partially schematic side views illustrating tilting of the transfer vessel **122** in accordance with embodiments of the present technology. It is appreciated that FIGS. **6**A-**6**C can represent tilting of the transfer vessel **122** in either the first arrangement of FIG. **4** or the second arrangement of FIG. **5**. Referring first to FIG. **6**A, the overhead crane **130** (not shown) can carry and place the transfer vessel **122** with molten metallics therein on the tilter **410**. A controller **612** can be operably coupled to the tilter **410** for controlling the tilting of the transfer vessel **122**. In some embodiments, operators and other personnel are at the left side of the granulator reactor **430** so that the overhead crane **130** does not cross paths with the personnel for safety reasons.
(47) Referring next to FIGS. **6**B and **6**C together, the tilter **410** can include a linear actuator **614** or other motorized component that can lift and tilt the transfer vessel **122**. Tilting the transfer vessel **122** allows the molten metallics therein to be transferred (e.g., poured) into the tundish **420**. The

controller **612** can operate the linear actuator **614** to tilt the transfer vessel **122** in a controlled manner such that, for example, the molten metallics flows out of the transfer vessel **122** and into the tundish **420** at a desired flow rate without overflowing the tundish or splashing too much molten metallics, which can result in material loss. As seen in FIG. **6**C, the tilter **410** can tilt the transfer vessel **122** by an angle greater than 90 degrees.

(48) FIGS. **7**, **8**A, and **8**B are front perspective, rear perspective, and side cross-sectional views, respectively, of the tundish **420** configured in accordance with embodiments of the present technology. The tundish **420** can include a tundish body **740**, a cover **748** disposed at least partially over the tundish body **740**, and an overflow channel **745** removably coupled to a front portion of the tundish body **740**. The tundish body **740** can define a cavity that can receive and pool molten metallics therein. More specifically, the cavity can have an open top **742** near the rear of the tundish body **740** that is not covered by the cover **748** and receives the stream of molten metallics from the transfer vessel **122**, and an outlet **743** at the lowest portion of and near a front end of the tundish body **740**. As best seen in FIG. **8**B, the tundish body **740** can have an angled bottom surface such that the molten metallics flows down toward the outlet **743**. In some embodiments, the tundish **420** further includes a nozzle **847** at the outlet **743** to ensure that the molten metallics flows out of the outlet **743** in a controlled manner. The nozzle **847** can comprise silica carbide, graphite, non-wetting materials, and/or other suitable material. In some embodiments, the nozzle **847** can be vibrating. The inner surface of the tundish body **740** defining the cavity can be lined with a liner material **747** such as refractory lining, silicas, aluminas, material not containing magnesium, and/or other suitable lining. For example, the lining material **1347** can be the same or a similar material as the sacrificial refractory lining included in the blast furnace **102**. A portion of the bottom surface of the tundish body **740** directly below the open top **742** can be covered with an impact pad **849** (FIG. **8**B) that can absorb the impact of the stream of molten metallics transferred (e.g., poured) from the transfer vessel **122**.

(49) In some embodiments, the tundish **420** additionally includes one or more flow control devices **845** (e.g., a ferrostatic head flow control device) coupled to sidewalls of the tundish body **740** and extending at least partially into the cavity (e.g., downward, upward, sideways). The flow control devices **845** can be a static structure or an adjustable structure whose position and/or orientation relative to the tundish body **740** can be controlled. In some embodiments, the flow control device **845** comprises a solid plate. In some embodiments, the flow control device **845** comprises a plate with one or more holes extending therethrough at one or more angles and/or arranged in an array. In some embodiments, the tundish **420** further includes a level sensor **749***a* mounted on a sidewall of the tundish body **740** and/or one or more load sensors **749***b* mounted on the bottom surface of the tundish body **740**. Furthermore, the tundish **420** can include a plurality of trunnions or lifting lugs **741** coupled to and extending outward from the sidewalls of the tundish body **740**.

(50) The cover **748**, shown in FIG. **7** but omitted in FIGS. **8**A and **8**B to avoid obscuring certain details of the tundish **420**, can be positioned to prevent molten metallics from splashing and spilling out of the tundish **420**, which can otherwise lead to significant material loss. As shown in FIG. **7**, the cover **748** can be shaped and sized to leave the open top **742** near the rear end of the tundish **420** exposed for receiving the stream of molten metallics from the transfer vessel **122**. The cover **748** can include refractory lining (illustrated with patterning in FIG. **7**) to also prevent other materials (e.g., splashed molten metallics or slag from the transfer vessel **122**) from entering the tundish **420** outside of the open top **742**. Towards the front end of the tundish **420**, the cover **748** can have a curved edge **749**, exposing a portion of the cavity directly above the outlet **743**. In some embodiments, a stopper rod assembly (illustrated in and described in further detail below with reference to FIGS. **9** and **10**) can be coupled to the front portion of the tundish **420** such that the stopper rod assembly can extend into the portion of the cavity exposed by the curved edge **749** and reach the outlet **743**. It will be appreciated that the cover **748** can have other shapes and/or sizes to provide spilling prevention while allowing the stopper rod assembly access to the outlet **743**.

(51) The overflow channel **745** can define an overflow outlet **746** through which excess material can flow out of the tundish **420**. As best seen in FIG. **8**B, the overflow outlet **746** can be positioned higher than the outlet **743** so that the molten metallics primarily flows out of the tundish **420** via the outlet **743**. The overflow channel **745** can be removably coupled to the tundish body **740** via, e.g., bolts and/or other fasteners. Advantageously, when the tundish **420** is transported to a cleaning/repair area, the overflow channel **745** can be detached from the tundish body **740** to facilitate handling. For example, the cleaning/repair area may include a machine that can flip the tundish **420** upside down, and detaching the overflow channel **745** can make it easier to place the tundish **420** on the machine.
(52) In operation, the open top **742** of the tundish **420** receives the stream of molten metallics flowing out of the transfer vessel **122**. The cover **748** can help ensure that only the stream of molten metallics enters the tundish **420** and can prevent splashes from spilling over the sides of the tundish body **740**. After the stream of molten metallics hits the impact pad **849**, the molten metallics can flow downward toward the outlet **743** and eventually pool in the cavity. The level sensor **749***a* can measure the surface level of the molten metallics and the load sensors **749***b* can measure the weight of the molten metallics in the cavity. In some embodiments, the readings from the level sensor **749***a* and/or the load sensors **749***b* are transmitted to the controller **612** (FIGS. **6**A-**6**C) so that the tilt angle of the transfer vessel **122** can be controlled to achieve a desired flow rate at any given time. For example, if the level sensor **749***a* and/or the load sensors **749***b* indicate that there is too much molten metallics, the controller **612** can reduce the tilt angle until, e.g., the level sensor **749***a* and/or the load sensors **749***b* indicate that a sufficient amount of molten metallics has flowed out through the outlet **743** and/or the overflow outlet **746** and the transfer vessel **122** can be tilted more.
(53) The flow control devices **845** can serve multiple functions. First, the flow control device **845** can contain agitation of the molten metallics at the rear side of the tundish **420**. As molten metallics is transferred (e.g., poured) from the transfer vessel **122**, the stream can cause splashing, waves, and other forms of turbulent flow at around the open top **742**. The flow control device **845** can act as a barrier that blocks the agitation from crossing over towards the outlet **743**. In some embodiments, the flow control device **845** is controllable to adjust a height thereof. As a result, the flow of molten metallics exiting the tundish **420** via the outlet **743** can be relatively calm and/or laminar. Second, the flow control device **845** can act as a barrier that blocks slag or other impurities floating on or near the surface of the molten metallics from crossing over towards the outlet **743**. The slag that builds up at the open top **742** can be skimmed off the surface or eventually directed out of the tundish **420** via the overflow outlet **746**. As a result, the flow of molten metallics exiting the tundish **420** via the outlet **743** can be relatively devoid of slag. Third, the flow control device **845** can act as a vortex breaker that can prevent or at least impede the formation of vortices in the molten metallics. It is appreciated that the tundish **420** can include a plurality of the flow control devices **845**, and different ones of the flow control devices **845** can have different shapes and/or dimensions, and/or extend in different directions to provide the various functions described herein.
(54) In the illustrated embodiment, the tundish **420** includes a total of four lifting lugs **741**, as best seen in FIGS. **7** and **8**A. The overhead crane **130** (FIG. **1**) can be operated to lift, lower, and/or transport the tundish **420** using the lifting lugs **741**. For example, the overhead crane **130** can be used to reposition the tundish **420** at a lower or higher height to increase or decrease the distance between the tundish **420** and other components of the granulator unit **150**. In another example, the overhead crane **130** can be used to remove the tundish **420** from the granulator unit **150** for, e.g., maintenance. Because the system **100** is a continuous system and there may not be a conventional “downtime” during which components may undergo maintenance on-site, the ability to quickly and easily transport the tundish **420** for maintenance or replacement can be important to ensure that the system **100** remains continuous.
(55) FIG. **9** is a perspective view of a stopper rod assembly **980** configured in accordance with

embodiments of the present technology. The stopper rod assembly **980** can include a base plate **982**, a vertical member **984** extending vertically from the base plate **982**, a horizontal member **986** extending horizontally from the vertical member **984**, a motor **985** operably coupled to the vertical member **984** and/or the horizontal member **986**, and a stopper rod **988** extending downward from a distal tip of the horizontal member **986**. The base plate **982** can be coupled to the tundish **420**, e.g., bolted to the tundish body **740**. The stopper rod **988** can be shaped and sized to fit in the outlet **743** and/or the nozzle **847** of the tundish **420**. In some embodiments, the stopper rod **988** includes refractory lining or other suitable lining to protect the stopper rod **988** from the hot molten metallics. In operation, a controller (not shown) can be used to operate the motor **985** to move the horizontal member **986**, and thus the stopper rod **988**, vertically along the length of the vertical member **984**. Therefore, the stopper rod **988** can be remotely controlled to selectively plug the outlet **743** of the tundish **420** and thereby control the flow of molten metallics out of the tundish **420**.

(56) FIG. **10** is a perspective view of a tundish **1040** and a stopper rod assembly **1050** assembled together and configured in accordance with embodiments of the present technology. It is appreciated that while FIG. **10** illustrates different embodiments of a tundish and stopper rod assembly than FIGS. **7-9**, the tundish **420** and the stopper rod assembly **980** illustrated in FIGS. **7-9** can be assembled together in a similar manner. In the illustrated embodiment, the tundish **1040** includes a tundish body **1041**, a cover **1048** disposed over the tundish body **1041**, and an overflow channel **1045** removably coupled to the front of the tundish body **1041**. The stopper rod assembly **1050** can include a mounting frame **1051**, an actuator **1056** (e.g., a linear actuator) secured to the mounting frame **1051**, a vertical member **1052** coupled to and extending upward from the actuator **1056**, a horizontal member **1054** extending horizontally from the upper tip of the vertical member **1052**, and a stopper rod **1058** coupled to and extending downward from the distal tip of the horizontal member **1054**. As shown, the mounting frame **1051** can be attached to an outer side wall of the tundish body **1041** via fasteners, brackets, or other coupling mechanisms. More specifically, the mounting frame **1051** can be secured at a position aligned with an outlet **1043** of the tundish **1040**. The horizontal member **1054** can extend over the sidewall of the tundish body **1041** such that the stopper rod **1058** hangs directly above the outlet **1043**.

(57) In operation, the actuator **1056** can move the vertical member **1052** between a raised position (illustrated in FIG. **10**) and a lowered position. When the vertical member **1052** is in the raised position, the stopper rod **1058** is spaced apart from the outlet **1043**, allowing molten metallics to flow out through the outlet **1043**. When the vertical member **1052** is in the lowered position, the stopper rod **1058** can at least partially plug the outlet **1043** to impede flow of molten metallics therethrough. One of ordinary skill in the art will appreciate that the actuator **1056** can be precisely controlled to adjust the position of the stopper rod **1058** to various heights between the raised and lowered positions to plug the outlet **1043** by varying degrees. Therefore, the actuator **1056** can be controlled to provide varying levels of flow control through the outlet **1043** of the tundish **1040**.

(58) Advantageously, attaching the stopper rod assembly **1050** directly to the tundish **1040** as opposed to, e.g., a frame structure supporting the tundish **1040**, can increase safety levels during operation. For example, if the stopper rod **1058** becomes stuck in the outlet **1043** or elsewhere, the tundish **1040** and the stopper rod assembly **1050** can be removed together for repair. If the stopper rod assembly **1050** were attached to another structure (e.g., a frame structure that cannot be easily removed from the on-site location), it can be difficult and unsafe to separate the tundish **1040** and the stopper rod assembly **1050** at the on-site location.

(59) FIG. **11** is a schematic cross-sectional view of the granulation reactor **430** configured in accordance with embodiments of the present technology. The granulation reactor **430** can include a reactor body **1160** and a spray head or target **1163** coupled to the reactor body **1160**. The reactor body **1160** can define a cavity **1166** therein and an overflow channel **1162** extending around the upper portion of the cavity **1166**. As illustrated, the cavity **1166** and the overflow channel **1162** can

be open at the top to receive material from above. More specifically, the cavity **1166** can be positioned to receive the molten metallics flowing out of the tundish **420** through the outlet **743** and the nozzle **847**. The reactor **430** can include an outlet **1168** of the cavity **1166** at the lower portion of the reactor body **1160**. The overflow channel **1162** can be positioned to receive overflow material (e.g., molten metallics mixed with slag) from the tundish **420**. For example, the lengths of the overflow channel **745** (of the tundish **420**) can be set so that overflow materials flow down into the overflow channel **1162** (of the granulation reactor **760**). The overflow materials received in the overflow channel **1162** can be removed via an outlet and sent to further processing. In some embodiments, the granulation reactor **430** further includes lifting lugs coupled to the reactor body **1160** for facilitating lifting of the granulation reactor **430** by the overhead crane **130**.

(60) The target **1163** can be secured at the center of the cavity **1166**. For example, in some embodiments, the target **1163** is secured via one or more struts extending from the reactor body **1160** (e.g., like a tripod). Cooled water can enter the cavity **1166** via the reactor body **1160** and be pooled and/or circulated therein.

(61) In operation, the granulation reactor **430** can continuously or intermittently receive cooled water from the cooling system **160** and at least partially fill the cavity **1166** with the cooled water. The volumetric capacity of the cavity **1166** can be between 10,000-100,0000 gallons or between 20,000-40,000 gallons. The molten metallics flowing down from the tundish **420** can impact the target **1163**. The target **1163** can be shaped and sized to spray the molten metallics into different directions. The molten metallics that enters the cooled water is cooled and becomes granulated. One of ordinary skill in the art will appreciate that the falling distance between the tundish **420** and the target **1163** can affect the shape, size, and quality of the resulting granulated products. As discussed above, the overhead crane **130** can adjust the height of the tundish **420** relative to the target **1163** to produce granulated products with desired properties (e.g., shape, size, quality). The formed granulated products can exit the granulation reactor **430** via the outlet **1168**.

(62) FIGS. **12**A and **12**B are schematic side and enlarged, side cross-sectional views, respectively, of the ejector **440** configured in accordance with embodiments of the present technology. Referring first to FIG. **12**A, the ejector **40** can include an inlet **1272** and a jet inlet **1273**, and the ejector **440** can be coupled to a lift line **1274** having a first rock box **1275***a*, a second rock box **1275***b*, and an outlet **1278**. The inlet **1272** can be positioned at the outlet **1168** of the granulation reactor **430** to receive the granulated products therefrom. In the illustrated embodiment, the inlet **1272** has a funnel shape generally corresponding to the shape of the lower portion of the granulation reactor **430**. The jet inlet **1273** can be coupled to receive a water jet stream and/or a stream of compressed air, which can provide enough pressure to push the granulated products up the lift line **1274**. In the illustrated embodiment, the ejector **440** and the lift line **1274** are mounted on rails **1276**, which allow the ejector **440** and the lift line **1274** to be easily removed from the granulator unit **150** for maintenance, contributing to the modularity of the granulator unit **150**.

(63) Referring next to FIG. **12**B, which illustrates an enlarged, side cross-sectional view of the first rock box **1275***a*, the first rock box **1275***a* can be internally connected to the lift line **1274** where the lift line **1274** changes angle. In other words, the first rock box **1275***a* serves as an extended corner space or dead zone for the lift line **1274**. In operation, as the granulated products from the granulation reactor **430** and the water and/or compressed air streams flow through the lift line **1274**, the first rock box **1275***a* can collect granules **1277**. The granules **1277** can remain in the corner of the first rock box **1275***a* by virtue of the fluid velocity and pressure. Thus, the first rock box **1275***a* can help reduce wear and tear on the lift line **1274** by managing the impact and abrasion caused by the granulated products changing direction, and the buildup of the granules **1277** can act as a buffer to absorb the impact of incoming material and reduce the velocity of the flow. In some embodiments, the lift line **1274** is lined with a liner material **1271** such as ceramic, silicon carbide, titanium, tungsten carbide, and/or other suitable material.

(64) FIG. **13** is a schematic side view of the dewatering assembly **460** configured in accordance

with embodiments of the present technology. The dewatering assembly **460** can include a frame structure **1381**, a dewatering screen **1384** supported on the frame structure **1381**, and an imaging device **1385** positioned to capture images of the products on the dewatering screen **1384**. The outlet **1278** of the lift line **1274** can couple to an inlet **1382** of the dewatering assembly **460** so that the dewatering screen **1384** can receive wet granulated products from the granulation reactor **430** via the ejector **440**. The dewatering assembly **460** can further include an outlet chute **1386** at the end of the dewatering screen **1384** so that the screened products can be collected separately.

(65) In operation, as the granulated products from the lift line **1274** move from the inlet **1382** to the outlet chute **1386**, the dewatering screen **1384** can filter out water and particles below a threshold size. The threshold size can be between 0.1-10 mm, such as about 0.1 mm, 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, or 10 mm. The filtered out particles can be directed to the classifier assembly **470** via a pipe **1388** underneath the dewatering screen **1384**, and the products that reach the outlet chute **1386** can form the GMU products. In some embodiments, the imaging device **1385** can be used to perform optical granulometry, which involves visually inspecting the size distribution of the particles. If the particles are generally smaller than expected or desired, this may be an indication that the flow rate of the molten metallics is too fast. Accordingly, the images taken by the imaging device **1385** can be used in a feedback loop with components of the system **100** that manage flow rate, such as (i) the controller **612** for controlling the tilt angle of the transfer vessel **122**, (ii) controllers for adjusting the position of the flow control devices **845** in the tundish **420**, (iii) the motor **985** or the actuator **1056** for adjusting the height of the stopper rod **988** or **1058**, and/or (iv) the overhead crane **130** for adjusting the height of the tundish **420** relative to the granulation reactor **430**.

(66) FIG. **14** is a schematic side view of the classifier assembly **470** configured in accordance with embodiments of the present technology. The classifier assembly **470** can include a housing **1492** having an inlet **1494** that can couple to the pipe **1388** and an outlet chute **1498**. In operation, the inlet **1494** can receive the slurry (e.g., water and GMU fines) from the dewatering assembly **460** via the pipe **1388**. The GMU fines can be separated and released through the outlet chute **1498** to be collected separately. The remaining water can be directed to, e.g., the return water sump **202**.

(67) FIG. **15** is a schematic process flow diagram illustrating granulation of metal in accordance with embodiments of the present technology. The transfer vessel **122** can transport molten metallics from the blast furnace **102** and/or the desulfurization unit **140** to the granulation unit **150** (see FIG. **1**). The controller **612** can operate the tilter **410** to control the tilt angle of the transfer vessel **122** to transfer (e.g., pour) the molten metallics from the transfer vessel **122** to the tundish **420** at a desired flow rate. The transfer vessel **122** can include a slide gate or valve with associated controls. In some embodiments, the granulator unit **150** further includes a trough, bucket, tray, or other collector positioned below the transfer vessel **122** and/or the tundish **420** to receive any molten metallics or other material that may spill. A fume hood **1502** can be positioned to collect emissions from the molten metallics flowing through the tundish **420**. The stopper rod assembly **980** can be coupled to the tundish **420** and operated to control the flow of molten metallics out of the tundish **420** and into the granulation reactor **430**.

(68) The granulation reactor **430** can receive cool water from a cold water supply. The molten metallics exiting the tundish **420** can impact a target of the granulation reactor **430** to be sprayed over the water pooled inside the granulation reactor **430**. The granulation reactor **430** can granulate the molten metallics to form granulated products, such as by cooling the molten metallics. The heated water can be sent to a tank, hot well pumps, and eventually return to the cooling system **160**. In some embodiments, a drain pump **1532** is included between the granulation reactor **430** and the tank for maintenance purposes. The ejector **440** can receive ejector water and/or compressed air to transfer the granulated products from the granulation reactor **430** to the dewatering assembly **460**. The dewatering assembly **460** can dry and filter (e.g., by size) the granulated products to output GMUs products. In some embodiments, the granulator units **150** are configured to produce GMUs

at a rate that matches an output rate of the blast furnace **102**. The filtrate from the dewatering assembly **460** can be sent to the classifier assembly **470**, which can sort out and output GMU fines. The classifier discharge (e.g., remaining water and particulates therein) can be directed to the return water sump **202** or other processing. The various components of the granulator units **150** can be powered electrically, hydraulically, and/or via other methods.

III. Methods for Producing GMUs

(69) FIG. **16** is a flowchart illustrating a method **1600** for producing GMUs at a liquid hot metal processing unit in accordance with embodiments of the present technology. While the steps of the method **1600** are described below in a particular order, one or more of the steps can be performed in a different order or omitted, and the method **1600** can include additional and/or alternative steps. Additionally, although the method **1600** may be described below with reference to the embodiments of the present technology described herein, the method **1600** can be performed with other embodiments of the present technology.

(70) The method **1600** begins at block **1602** by transferring molten metallics to a ladle in the liquid hot metal processing unit. In some embodiments, transferring comprises operating a torpedo car to transfer the molten metallics from a blast furnace external to the liquid hot metal processing unit to the ladle in the liquid hot metal processing unit.

(71) At block **1604**, the method **1600** continues by transporting the ladle to a granulator unit in the liquid hot metal processing unit. The granulator unit can include a tilter positioned to receive and tilt the ladle. In some embodiments, the liquid hot metal processing unit includes a plurality of granulator units and transporting can comprise transporting the ladle to one of the plurality of granulator units. In some embodiments, transporting comprises operating an overhead crane in the liquid hot metal processing unit to transport the ladle to the granulator unit.

(72) At block **1606**, the method **1600** continues by tilting, using the tilter, the ladle to transfer the molten metallics to a tundish of the granulator unit. For example, the tilter can include a linear actuator, and a controller operably coupled to the linear actuator can be used to tilt the ladle in a controlled manner to achieve a desired flow rate of molten metallics to the tundish.

(73) At block **1608**, the method **1600** continues by directing the molten metallics from the tundish into a reactor of the granulator unit. In some embodiments, the method **1600** further comprises moving (e.g., dithering) a stopper rod to control a flow rate of the molten metallics out of an outlet of the tundish.

(74) At block **1610**, the method **1600** continues by granulating the molten metallics in the reactor to form granulated metallic units (GMUs). For example, the reactor can hold cooled water and the molten metallics can be dropped, poured, sprayed, or otherwise transferred into the water so that the molten metallics can cool and become GMUs.

(75) In some embodiments, prior to block **1602**, the method **1600** further comprises feeding the molten metallics to a BOF vessel in the liquid hot metal processing unit, and delivering oxygen to the molten metallics in the BOF vessel. The molten metallics can subsequently be transferred from the BOF vessel to the ladle. Delivering oxygen can comprise heating and/or reducing a carbon content of the molten metallics in the BOF vessel. In some embodiments, the method **1600** further comprises reducing a sulfur content of the molten metallics, such as by adding at least one of calcium carbide or magnesium to the molten metallics.

IV. Examples

(76) The present technology is illustrated, for example, according to various aspects described below as numbered examples (1, 2, 3, etc.) for convenience. These are provided as examples and do not limit the present technology. It is noted that any of the dependent examples may be combined in any combination, and placed into a respective independent example. The other examples can be presented in a similar manner. 1. A liquid hot metal processing system for producing granulated metallic units (GMUs), the system comprising: a liquid hot metal processing unit including: a granulator unit including: a tilter positioned to receive a ladle, a controller

operably coupled to the tilter to control a tilt of the ladle, a tundish positioned to receive molten metallics from the ladle, and a reactor positioned to receive the molten metallics from the tundish, wherein the reactor is configured to cool the molten metallics and form GMUs. 2. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes: the ladle configured to receive and store the molten metal; and an overhead crane configured to transfer the ladle to and from the tilter. 3. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes: a basic oxygen furnace (BOF) vessel configured to receive the molten metal; and an oxygen lance insertable into the BOF vessel to deliver oxygen gas to the molten metal. 4. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes an oxygen lance configured to deliver oxygen gas to the molten metal, and wherein the oxygen gas is configured to react with carbon in or on the molten metallics to heat the molten metal. 5. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes an oxygen lance configured to deliver oxygen gas to the molten metal, and wherein the oxygen gas is configured to react with carbon in or on the molten metallics to reduce a carbon content of the molten metal. 6. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes a desulfurization unit configured to reduce a sulfur content of the molten metal. 7. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes a desulfurization unit configured to reduce a sulfur content of the molten metal, and wherein the desulfurization unit is configured to reduce the sulfur content of the molten metallics by providing at least one of calcium carbide or magnesium to the molten metal. 8. The system of any of the examples herein, wherein the ladle comprises a first ladle, wherein the granulator unit comprises a first granulator unit, and wherein the liquid hot metal processing unit further includes a second ladle and a second granulator unit positioned adjacent to the first granulator unit, wherein the second granulator unit includes: a second tilter positioned to receive and tilt the second ladle; a second controller operably coupled to the second tilter to control tilting of the second ladle; a second tundish positioned to receive the molten metallics from the second ladle; and a second reactor positioned to receive the molten metallics from the second tundish, wherein the second reactor is configured to cool the molten metallics to form granulated metallic units (GMUs). 9. The system of any of the examples herein, wherein the granulator unit further includes a stopper rod assembly coupled to the tundish, wherein the stopper rod assembly include a stopper rod and an actuator operably coupled to move the stopper rod into and out of an outlet of the tundish. 10. The system of any of the examples herein, wherein the granulator unit further includes an ejector positioned to receive the GMUs from the reactor and a lift line downstream of the reactor, wherein lift line includes a curved region, and wherein the ejector further includes a rock box at the curved region, wherein the rock box is configured to receive and store a portion of the GMUs received in the ejector. 11. The system of any of the examples herein, wherein the granulator unit further includes an ejector positioned to receive the GMUs from the reactor and a lift line downstream of the reactor, wherein an inner surface of the lift line is lined with a liner material comprising ceramic, silicon carbide, titanium, and/or tungsten carbide. 12. The system of any of the examples herein, wherein the granulator unit further includes a dewatering assembly positioned downstream of the reactor, wherein the dewatering assembly is configured to dry the GMUs and filter the GMUs by size. 13. The system of any of the examples herein, wherein the dewatering assembly is configured to filter out GMU fines less than 1 millimeter in size. 14. The system of any of the examples herein, wherein the granulator unit further includes an imaging device positioned to capture images of the GMUs on the dewatering assembly, wherein the images captured by the imaging device are configured to be used in an optical granulometry feedback system to adjust a flow rate of the molten metallics into the reactor. 15. The system of any of the examples herein, wherein the granulator unit further includes a classifier assembly positioned downstream of the dewatering assembly, wherein the classifier assembly is configured to classify filtrate received from the dewatering assembly and output GMUs fines. 16. The system of any of the examples herein, wherein the granulator unit further includes: a dewatering assembly positioned downstream of the reactor; and a classifier assembly positioned downstream of the dewatering assembly, wherein the overhead crane is further configured to selectively and individually lift the tundish, the reactor, the dewatering assembly, or the classifier assembly. 17. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes a dust collection unit, and wherein the granulator unit further includes a fume hood positioned to collect emissions from the tundish and direct the emissions to the dust collection unit. 18. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes a transfer vessel preparation unit configured to deslag and dekish the ladle. 19. The system of any of the examples herein, wherein the liquid hot metal processing unit further includes a cooling system configured to provide cooled water to the reactor. 20. The system of any of the examples herein, further comprising: a blast furnace configured to melt iron ore to output the molten metal; and a torpedo car configured to transport the molten metallics from the blast furnace to the liquid hot metal processing unit. 21. A method for producing granulated metallic units (GMUs) at a liquid hot metal processing unit, the method comprising: transferring molten metallics to a ladle in the liquid hot metal processing unit; transporting the ladle to a granulator unit in the liquid hot metal processing unit, wherein the granulator unit includes a tilter positioned to receive and tilt the ladle; tilting the ladle via the tilter to transfer the molten metallics to a tundish of the granulator unit; directing the molten metallics from the tundish into a reactor of the granulator unit; and granulating the molten metallics in the reactor to form GMUs. 22. The method of any of the examples herein, further comprising: feeding the molten metallics to a basic oxygen furnace (BOF) vessel in the liquid hot metal processing unit; and delivering oxygen to the molten metallics in the BOF vessel to form decarbonated molten metal, wherein transferring comprises transferring the decarbonated molten metallics from the BOF vessel to the ladle. 23. The method of any of the examples herein, wherein delivering oxygen comprises heating the molten metallics in the BOF vessel. 24. The method of any of the examples herein, wherein delivering oxygen comprises reducing a carbon content of the molten metallics in the BOF vessel. 25. The method of any of the examples herein, wherein transporting comprises transporting the ladle to one of a plurality of granulator units in the liquid hot metal processing unit. 26. The method of any of the examples herein, further comprising reducing a sulfur content of the molten metal. 27. The method of any of the examples herein, wherein reducing the sulfur content comprises adding at least one of calcium carbide or magnesium to the molten metal. 28. The method of any of the examples herein, further comprising moving a stopper rod to control a flow rate of the molten metallics out of an outlet of the tundish. 29. The method of any of the examples herein, wherein transporting comprises operating an overhead crane in the liquid hot metal processing unit to transport the ladle to the granulator unit. 30. The method of any of the examples herein, wherein transferring comprises operating a torpedo car to transfer the molten metallics from a blast furnace external to the liquid hot metal processing unit to the ladle in the liquid hot metal processing unit.

V. Conclusion

(77) It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the present disclosure. In some cases, well known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments of the present technology. Although steps of methods may be presented herein in a particular order, alternative embodiments may perform the steps in a different order. Similarly, certain aspects of the present technology disclosed in the context of particular embodiments can be combined or eliminated in other embodiments. Furthermore, while advantages associated with certain embodiments of the present technology may have been disclosed in the context of those embodiments, other embodiments can also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages or other advantages disclosed herein to fall within the scope of the technology.

Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein, and the invention is not limited except as by the appended claims.

(78) Throughout this disclosure, the singular terms “a,” “an,” and “the” include plural referents unless the context clearly indicates otherwise. Additionally, the term “comprising,” “including,” and “having” should be interpreted to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded.

(79) Reference herein to “one embodiment,” “an embodiment,” “some embodiments” or similar formulations means that a particular feature, structure, operation, or characteristic described in connection with the embodiment can be included in at least one embodiment of the present technology. Thus, the appearances of such phrases or formulations herein are not necessarily all referring to the same embodiment. Furthermore, various particular features, structures, operations, or characteristics may be combined in any suitable manner in one or more embodiments.

(80) Unless otherwise indicated, all numbers expressing concentrations, shear strength, and other numerical values used in the specification and claims, are to be understood as being modified in all instances by the term “about.” “About” as used herein can represent a range of plus or minus 10% of the stated value. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present technology. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Additionally, all ranges disclosed herein are to be understood to encompass any and all subranges subsumed therein. For example, a range of “1 to 10” includes any and all subranges between (and including) the minimum value of 1 and the maximum value of 10, i.e., any and all subranges having a minimum value of equal to or greater than 1 and a maximum value of equal to or less than 10, e.g., 5.5 to 10.

(81) The disclosure set forth above is not to be interpreted as reflecting an intention that any claim requires more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.

Claims

1. A liquid hot metal processing system for producing granulated metallic units (GMUs), the system comprising: a liquid hot metal processing unit including: a granulator unit including: a tilter positioned to receive a ladle, a controller operably coupled to the tilter to control a tilt of the ladle, a tundish positioned to receive molten metallics from the ladle, and a reactor positioned to receive the molten metallics from the tundish, wherein the reactor is configured to cool the molten metallics and form the GMUs, an ejector positioned to receive GMUs from the reactor, and a lift line downstream of the reactor, wherein the lift line includes a rock box configured to receive and store a portion of the GMUs received by the ejector.

2. The system of claim 1, wherein the liquid hot metal processing unit further includes: the ladle configured to receive and store the molten metallics; and an overhead crane configured to transfer the ladle to and from the tilter.

3. The system of claim 1, wherein the liquid hot metal processing unit further includes: a basic oxygen furnace (BOF) vessel configured to receive the molten metallics; and an oxygen lance insertable into the BOF vessel to deliver oxygen gas to the molten metallics.

4. The system of claim 1, wherein the liquid hot metal processing unit further includes an oxygen lance configured to deliver oxygen gas to the molten metallics, and wherein the oxygen gas is configured to react with carbon in or on the molten metallics to reduce a carbon content of the molten metallics.
5. The system of claim 1, wherein the liquid hot metal processing unit further includes a desulfurization unit configured to reduce a sulfur content of the molten metallics, and wherein the desulfurization unit is configured to reduce the sulfur content of the molten metallics by providing at least one of calcium carbide or magnesium to the molten metallics.
6. The system of claim 1, wherein the ladle comprises a first ladle, wherein the granulator unit comprises a first granulator unit, and wherein the liquid hot metal processing unit further includes a second ladle and a second granulator unit positioned adjacent to the first granulator unit, wherein the second granulator unit includes: a second tilter positioned to receive and tilt the second ladle; a second controller operably coupled to the second tilter to control tilting of the second ladle; a second tundish positioned to receive the molten metallics from the second ladle; and a second reactor positioned to receive the molten metallics from the second tundish, wherein the second reactor is configured to cool the molten metallics to form granulated metallic units (GMUs).
7. The system of claim 1, wherein the granulator unit further includes a stopper rod assembly coupled to the tundish, wherein the stopper rod assembly include a stopper rod and an actuator operably coupled to move the stopper rod into and out of an outlet of the tundish.
8. The system of claim 1, wherein the lift line includes a curved region, and wherein the rock box is at the curved region.
9. The system of claim 1, wherein the granulator unit further includes a dewatering assembly positioned downstream of the reactor, wherein the dewatering assembly is configured to filter out GMU fines less than 1 millimeter in size.
10. The system of claim 1, wherein the granulator unit further includes a dewatering assembly positioned downstream of the reactor, wherein the granulator unit further includes an imaging device positioned to capture images of the GMUs on the dewatering assembly, wherein the images captured by the imaging device are configured to be used in an optical granulometry feedback system to adjust a flow rate of the molten metallics into the reactor.
11. The system of claim 1, wherein the granulator unit further includes a dewatering assembly positioned downstream of the reactor, wherein the granulator unit further includes a classifier assembly positioned downstream of the dewatering assembly, wherein the classifier assembly is configured to classify filtrate received from the dewatering assembly and output GMUs fines.
12. The system of claim 1, wherein the liquid hot metal processing unit further includes a transfer vessel preparation unit configured to deslag and dekish the ladle.
13. The system of claim 1, further comprising: a blast furnace configured to melt iron ore to output the molten metallics; and a torpedo car configured to transport the molten metallics from the blast furnace to the liquid hot metal processing unit.